LONG FLEX CIRCUITS

Solutions for Medical Applications

RoHS COMPLIANT





Tech Etch long flex circuits support lengths up to 60 inches for space-constrained, motion-driven medical applications requiring ultra-thin, lightweight interconnects. Process-validated through Laser Direct Imaging, etch, and plating, these builds accommodate fine line/space geometries, ultra-thin core constructions, and 9 µm metal layers with precise layer-to-layer alignment. Optimized for catheter-based systems, surgical instrumentation, imaging devices, and wearable platforms, they provide stable electrical performance and mechanical reliability under dynamic flex conditions.

MEDICAL APPLICATIONS + LIFE SAVING PRECISION

WEARABLE MEDICAL

- Full-body biosensors embedded in garments or adhesive patches
- Post-surgical monitoring systems covering wide anatomical zones

IMAGING EQUIPMENT

- Interconnects for ultrasound probes and imaging consoles
- Enhanced Flex Circuits for MRI and CT scan systems

CATHETER BASED TOOLS

- Embedded circuits in electro-physiology and ablation catheters
- Real-time signal routing from endoscopic tools to processors

SURGICAL ROBOTICS

- Power/signal delivery through articulating arms and instrument tracks
- Withstands repetitive motion and sterilization demands

WHY TECH ETCH LONG FLEX CIRCUITS?

Supports minimally invasive, space-constrained designs	Maintains signal integrity across dynamic movements	Lightweight, low-profile alternative to wire harnesses	ISO 13485 certified for medical device manufacturing

ATTRIBUTE	PREFERRED	ADVANCED
Lead Width (min)	0.003"	0.002"
Lead Spacing (min)	0.003"	0.002"
Flex Length (max)	40"	60"

REQUEST A QUOTE

Request a fast quote for Long Flex Circuits. Requests for standard product will be answered within 24 hours.

INNOVATING A BETTER WORLD, ONE PART AT A TIME.

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